

8. To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

Code:

```

GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_INPUT 300
#define MAX_OUTPUT 500

/* Expand a single variable */
void expandVariable(const char *variable, char *value) {
    const char *result = getenv(variable);
    if (result != NULL) {
        strcpy(value, result);
    }
    else {
        /* Undefined variables become empty */
        value[0] = '\0';
    }
}

/* Expand variables inside a string */
void expandVariables(const char *input, char *output) {
    int i = 0;
    int j = 0;

    while (input[i] != '\0' && j < MAX_OUTPUT - 1) {
        /* Detect variable reference */
        if (input[i] == '$') {
            i++;
            char variable[100];
            int k = 0;

            /* Handle ${VARIABLE} format */
            if (input[i] == '{') {
                i++;
                while (input[i] != '\0' &&
                    input[i] != '}' &&
                    k < 99) {
                    variable[k++] = input[i++];
                }
                variable[k] = '\0';
                /* Skip closing brace */
                if (input[i] == '}')
                    i++;
            }
            /* Handle $VARIABLE format */
            else {

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                while (input[i] != '\0' &&
                    (isalnum((unsigned char)input[i]) ||
                     input[i] == '-' &&
                     k < 99)) {
                    variable[k++] = input[i++];
                }
                variable[k] = '\0';
            }
            /* Expand variable */
            char value[200];
            expandVariable(variable, value);
            /* Copy expanded value */
            for (int x = 0;
                value[x] != '\0' &&
                j < MAX_OUTPUT - 1;
                x++) {
                output[j++] = value[x];
            }
        }
        else {
            output[j++] = input[i++];
        }
    }
    output[j] = '\0';
}

/* Display result */
void displayResult(const char *original,
                  const char *expanded) {
    printf("\nVariable Expansion Result\n");
    printf("-----\n");
    printf("Original : %s\n", original);
    printf("Expanded : %s\n", expanded);
    printf("-----\n");
}

int main() {
    char input[MAX_INPUT];
    char output[MAX_OUTPUT];
    printf("Variable Expansion Parser\n");
    printf("-----\n");

```

```

        else {
            output[j++] = input[i++];
        }
    }
    output[j] = '\0';
}

/* Display result */
void displayResult(const char *original,
                  const char *expanded) {
    printf("\nVariable Expansion Result\n");
    printf("-----\n");
    printf("Original : %s\n", original);
    printf("Expanded : %s\n", expanded);
    printf("-----\n");
}

int main() {
    char input[MAX_INPUT];
    char output[MAX_OUTPUT];

    printf("Variable Expansion Parser\n");
    printf("-----\n");

    /*
     * Set sample variables for testing.
     */
    setenv("NAME", "Ketana", 1);
    setenv("CITY", "Hyderabad", 1);
    setenv("COURSE", "CSE", 1);

    printf("Available variables:\n");
    printf("NAME = %s\n", getenv("NAME"));
    printf("CITY = %s\n", getenv("CITY"));
    printf("COURSE = %s\n", getenv("COURSE"));

    printf("Enter input: ");

    fgets(input, sizeof(input), stdin);

    /* Remove newline */
    input[strcspn(input, "\n")] = '\0';

    /* Expand variables */
    expandVariables(input, output);

    /* Display result */
    displayResult(input, output);

    return 0;
}

```

Result:

```

abhinav@Abhinav: ~/ x + v
abhinav@Abhinav: ~/SKILL$ nano Skill8a.c
abhinav@Abhinav: ~/SKILL$ gcc Skill8a.c -o s8a
abhinav@Abhinav: ~/SKILL$ ./s8a
Variable Expansion Parser
-----
Available variables:
NAME = Ketana
CITY = Hyderabad
COURSE = CSE

Enter input: Hello $NAME

Variable Expansion Result
-----
Original : Hello $NAME
Expanded : Hello Ketana
-----
abhinav@Abhinav: ~/SKILL$ ./s8a
Variable Expansion Parser
-----
Available variables:
NAME = Ketana
CITY = Hyderabad
COURSE = CSE

Enter input: I am $NAME from $CITY and I study $COURSE

Variable Expansion Result
-----
Original : I am $NAME from $CITY and I study $COURSE
Expanded : I am Ketana from Hyderabad and I study CSE
-----
abhinav@Abhinav: ~/SKILL$

```

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define MAX_INPUT 200
#define MAX_ARGS 20

/* Function pointer type for built-in commands */
typedef int (*BuiltinFunction)(char **args);

/* Shell state */
char current_directory[MAX_INPUT];

/* ----- BUILT-IN COMMANDS ----- */

/* cd command */
int builtin_cd(char **args) {
    if (args[1] == NULL) {
        printf("cd: missing argument\n");
        return 1;
    }

    if (chdir(args[1]) != 0) {
        perror("cd");
        return 1;
    }

    getcwd(current_directory, sizeof(current_directory));
    printf("Directory changed to: %s\n", current_directory);
    return 1;
}

/* pwd command */
int builtin_pwd(char **args) {
    (void)args;
    getcwd(current_directory, sizeof(current_directory));
    printf("%s\n", current_directory);
    return 1;
}

/* echo command */
int builtin_echo(char **args) {
    for (int i = 1; args[i] != NULL; i++) {
        printf("%s", args[i]);
        if (args[i + 1] != NULL)
            printf(" ");
    }
    printf("\n");
}

/* echo command */
int builtin_echo(char **args) {
    for (int i = 1; args[i] != NULL; i++) {
        printf("%s", args[i]);
        if (args[i + 1] != NULL)
            printf(" ");
    }
    printf("\n");
    return 1;
}

/* help command */
int builtin_help(char **args) {
    (void)args;
    printf("\nBuilt-in Commands:\n");
    printf("cd <directory>    - Change directory\n");
    printf("pwd                - Display current directory\n");
    printf("echo <text>         - Display text\n");
    printf("help               - Display help\n");
    printf("exit               - Exit shell\n");
    return 1;
}

/* exit command */
int builtin_exit(char **args) {
    (void)args;
    printf("Exiting shell...\n");
    return 0;
}

/* ----- DISPATCH TABLE ----- */
typedef struct {
    char *name;
    BuiltinFunction function;
} Builtin;

/* Dispatch table */
Builtin dispatchTable[] = {
    {"cd", builtin_cd},
    {"pwd", builtin_pwd},
    {"echo", builtin_echo},
    {"help", builtin_help},
    {"exit", builtin_exit}
};

int builtinCount =
    sizeof(dispatchTable) / sizeof(dispatchTable[0]);

/* ----- PARSE INPUT ----- */
```

```

GNU nano 8.7.1
/* ----- PARSE INPUT ----- */
int parseInput(char *input, char **args) {
    int count = 0;
    char *token = strtok(input, " \\t\\n");
    while (token != NULL && count < MAX_ARGS - 1) {
        args[count++] = token;
        token = strtok(NULL, " \\t\\n");
    }
    args[count] = NULL;
    return count;
}

/* ----- COMMAND DISPATCH ----- */
int executeBuiltin(char **args) {
    if (args[0] == NULL)
        return 1;

    for (int i = 0; i < builtinCount; i++) {
        if (strcmp(args[0], dispatchTable[i].name) == 0) {
            /* Execute function from dispatch table */
            return dispatchTable[i].function(args);
        }
    }

    printf("Invalid command: %s\\n", args[0]);
    printf("Type 'help' to see available commands.\\n");
    return 1;
}

/* ----- MAIN ----- */
int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    getcwd(current_directory, sizeof(current_directory));

    printf("Built-in Command Shell\\n");
    printf("-----\\n");
    printf("Type 'help' for available commands.\\n\\n");

    while (1) {
        printf("shell[%s]> ", current_directory);
        fflush(stdout);

```

```

    }

    /* ----- MAIN ----- */
    int main() {
        char input[MAX_INPUT];
        char *args[MAX_ARGS];

        getcwd(current_directory, sizeof(current_directory));

        printf("Built-in Command Shell\\n");
        printf("-----\\n");
        printf("Type 'help' for available commands.\\n\\n");

        while (1) {
            printf("shell[%s]> ", current_directory);
            fflush(stdout);

            if (fgets(input, sizeof(input), stdin) == NULL) {
                break;
            }

            /* Remove newline */
            input[strcspn(input, "\\n")] = '\\0';

            /* Ignore empty input */
            if (strlen(input) == 0)
                continue;

            /* Convert input into arguments */
            parseInput(input, args);

            /* Dispatch command */
            int result = executeBuiltin(args);

            /* Exit when command returns 0 */
            if (result == 0)
                break;
        }

        return 0;
    }
}

```

Result:

```
abhinav@Abhinav:
abhinav@Abhinav: ~/SKILL$ nano Skill8b.c
abhinav@Abhinav: ~/SKILL$ gcc Skill8b.c -o s8b
abhinav@Abhinav: ~/SKILL$ ./s8b
Built-in Command Shell
-----
Type 'help' for available commands.

shell[/home/abhinav/SKILL]> pwd
/home/abhinav/SKILL
shell[/home/abhinav/SKILL]> echo Hello World
Hello World
shell[/home/abhinav/SKILL]> help

Built-in Commands:
  cd <directory> - Change directory
  pwd           - Display current directory
  echo <text>    - Display text
  help          - Display help
  exit          - Exit shell
shell[/home/abhinav/SKILL]> cd ..
Directory changed to: /home/abhinav
shell[/home/abhinav]> pwd
/home/abhinav
shell[/home/abhinav]> invalid
Invalid command: invalid
Type 'help' to see available commands.
shell[/home/abhinav]> exit
Exiting shell...
abhinav@Abhinav: ~/SKILL$
```